

# Scalable dissipative quantum error correction for discrete-variable codes

Ivan Rojko<sup>1,2</sup>, Elias Zapusek<sup>1,2</sup> and Florentin Reiter<sup>1,2,3</sup>

<sup>1</sup>*Institute for Quantum Electronics, ETH Zürich, Otto-Stern-Weg 1, 8093 Zürich, Switzerland*

<sup>2</sup>*Quantum Center, ETH Zürich, 8093 Zürich, Switzerland*

<sup>3</sup>*Fraunhofer Institute for Applied Solid State Physics IAF, Tullastr. 72, 79108 Freiburg, Germany*  
(Dated: July 18, 2025)

Dissipative quantum error correction (QEC) autonomously protects quantum information using engineered dissipation and offers a promising alternative to error correction via measurement and feedback. However, scalability remains a challenge, as correcting high-weight errors typically requires increasing dissipation rates and exponentially many correction operators. Here, we present a scalable dissipative QEC protocol for discrete-variable codes, correcting multi-qubit errors via a trickle-down mechanism that sequentially reduces errors weight. Our construction exploits redundancy in the Knill–Laflamme conditions to design correction operators that act on multiple error subspaces simultaneously, thereby reducing the overhead from exponential to polynomial in the number of required operators. We illustrate our approach with repetition codes under biased noise, showing a fourfold improvement in the exponential suppression factor at realistic physical error rates. Our approach connects autonomous QEC for discrete-variable codes with demonstrated dissipative protocols for bosonic codes and opens up new avenues for traditional measurement-feedback QEC and fault-tolerant quantum operations.

## Introduction

Quantum error correction (QEC) is an essential ingredient for fault tolerant quantum computation and drives advances in quantum metrology [1], simulation [2], and communication [3]. Standard QEC techniques rely on classical control theory, requiring auxiliary quantum systems, measurements of error syndromes, and feedback-conditioned operations to correct errors in the target system. Although recent experiments have demonstrated the feasibility of these QEC schemes [4–6] and achieved logical lifetimes greater than of best physical qubits in the system (known as break-even point) [7–12], they often rely on correction in post-processing or show limited advantage over uncorrected quantum systems. The main limitations arise from errors introduced during syndrome measurements [13–16] and the time delays caused by feedback loops [17–19].

Dissipative QEC [20–23], also known as autonomous QEC [24], represents an alternative paradigm in the context of protection of quantum information. These schemes replace measurements and standard error recovery techniques with coherent operations and natural dissipative channels. Numerous dissipative error correction schemes have been proposed and demonstrated in recent research. The most notable advancement have been observed in bosonic codes which encode and protect quantum information in systems with extended Hilbert space (e.g., quantum harmonic oscillators and qudit systems). In this context, dissipative correction schemes have been demonstrated for Gottesman-Kitaev-Preskill (GKP) [25], Cat [26], and Binomial [27] codes using both discrete- [13, 18, 28] and continuous-time [29–34] QEC approaches. Notably, break-even was first achieved using bosonic dissipative QEC [35, 36]. Even though these codes entail a low hardware overhead for encoding and correcting log-

ical information, their scalability through concatenation is still under active investigation [37].

Discrete-variable codes, i.e. those relying on a redundant use of two-level systems, represent another class of error correcting codes for which scalable architectures have recently been demonstrated using various technologies such as trapped ions, neutral atoms, superconducting circuits [4–6]. Various dissipative QEC schemes for such codes have been proposed [38–46] and implemented [47, 48]. The main challenge for these codes is however their scalability and in particular the ability to correct high-weight errors, where the weight denotes the number of elementary errors (typically single-qubit) separating the state from the logical subspace. Although recent studies [24, 49] suggest that dissipative QEC is, in principle, possible and yields favorable scaling of logical error rates with increasing code size, existing constructions require the engineered dissipation rate to scale polynomially with the code size and rely on an exponentially growing number of correction operators. As a result, until now, the existence of scalable dissipative schemes remains an open question.

In this work, we present a scalable construction of dissipative QEC schemes for discrete-variable systems, designed to correct high-weight errors that go beyond single-qubit noise. Our schemes operate in a *trickle-down* fashion: the system is driven through a sequence of lower-weight error states until it reaches the codespace. This construction is made possible by our observation that the Knill–Laflamme conditions [50], which typically relate the codespace to individual error subspaces, also apply between different error subspaces themselves. If two such subspace pairs satisfy the conditions identically, then the same operator that corrects one pair can also correct the other. This enables the design of correction operators

that simultaneously reduce the error weight of multiple subspaces simultaneously, significantly lowering the experimental overhead. Importantly, we show that this overhead reduction can be exponential, with the number of required correction operators scaling only polynomially with the code size. Given that many quantum codes are built from repetition codes, we illustrate our approach using a scalable dissipative correction scheme for repetition codes in biased noise environments [51], and demonstrate that the trickle-down mechanism enhances correction performance compared to standard dissipative QEC methods. It enhances the exponential error suppression factor by  $\sim 4$  times for a physical error rate of  $10^{-2}$ , thereby reducing the qubit overhead needed to reach a target logical error rate. Finally, we relate our approach to measurement-feedback QEC methods and to the recent developments in bosonic codes. This advancement offers insights into autonomously error-protected quantum sensors, memories and processing units.

## Results

### Dissipative QEC

We consider the problem of protecting a logical subspace  $\mathcal{C}$  of a Hilbert space  $\mathcal{H}$  from incoherent noise that disturbs the system. We model the dynamics responsible for the errors and their correction using a Lindblad equation

$$\dot{\rho} = (\mathcal{L}_e + \mathcal{L}_c)(\rho), \quad (1)$$

where both Liouvillians describe purely incoherent processes

$$\mathcal{L}_\bullet(\rho) := \sum_i \mathcal{D}[L_{\bullet,i}](\rho) \quad \text{for } \bullet \in \{e, c\}, \quad (2)$$

with  $\mathcal{D}[L](\rho) = L\rho L^\dagger - \frac{1}{2}(L^\dagger L\rho + \rho L^\dagger L)$ . The set of jump operators  $\{L_{e,i}\}$  describes the errors caused by undesired couplings of the system to the environment, while the set  $\{L_{c,i}\}$  represents the correction operators. The goal of the dissipative QEC is to carefully engineer the correction operations in such a way that the logical subspace  $\mathcal{C}$  is stabilized under the action of the combined Liouvillian  $\mathcal{L}_{\text{eff}} = \mathcal{L}_e + \mathcal{L}_c$ .

The form of the correction operators is determined by the encoding of the logical subspace  $\mathcal{C}$  into the physical Hilbert space  $\mathcal{H}$ . To be more specific, we consider the system to be composed of  $n$  qubits, i.e.  $\mathcal{H} = \mathbb{C}^{2^n}$ , and assume the code  $\mathcal{C}$  to be  $k$ -dimensional spanned by the logical states  $\{|\mu_L\rangle\}_{\mu=0}^{k-1}$ . The code choice is not unique and depends on the error set  $\mathbb{E} = \{E_i\}$  against which we want it to be robust. Note that each  $E_i$  corresponds to the accumulated error operator defined by a product of operators drawn from  $\mathbb{L} = \{L_{e,i}\} \cup \{L_{c,i}^\dagger\}$ . To be able to correct these errors, the code  $\mathcal{C}$  must satisfy the Knill–Laflamme condition  $P_C E_j^\dagger E_i P_C = \alpha_{ij} P_C \forall i, j$  with  $\alpha_{ij}$  being a scalar and  $P_C$  the projector on the logical subspace [50]. This condition ensures that there exist a par-

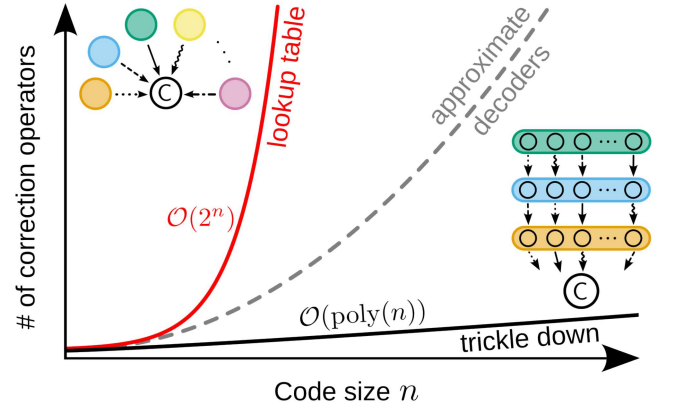


FIG. 1: *Number of correction operators vs. code size.* Schematic comparison of the number of jump operators required to correct an  $n$ -qubit code  $\mathcal{C}$  using lookup-table (red), trickle-down (black), and approximate decoder (dashed) schemes. The lookup-table assigns one operator per error subspace, scaling as  $\mathcal{O}(2^n)$ . The trickle-down approach groups subspaces by error weight and jump operators are designed to lower the error weight of several subspaces at once, yielding polynomial scaling (see Fig. 2(a) for details). The approximate decoder offers an intermediate solution.

tioning of the Hilbert space  $\mathcal{H}$  into subspaces, each associated with an individual invertible (or approximately invertible [52]) error process  $\tilde{E}_i$ . These new error operators are obtained by unitarily transforming the set  $\{E_i\}$  such that  $\alpha_{ij} = p_i \delta_{ij}$  with  $p_i$  quantifying the error detection probability of  $\tilde{E}_i$ . Without loss of generality, we consider  $\tilde{E}_i \equiv E_i$ . The recovery operation for the code  $\mathcal{C}$  can then be written as a completely positive trace preserving map

$$\mathcal{R}(\rho) = \sum_i U_i^\dagger P_i \rho P_i U_i + P_E \rho P_E \quad (3)$$

where  $P_i = U_i P_C U_i^\dagger \equiv E_i P_C U_i^\dagger / \sqrt{p_i}$  and  $P_E = \mathbb{1} - \sum_i P_i$  are the projectors onto the error subspace associated with  $E_i$  and onto the subspace of errors violating the correctability condition, respectively. The unitaries  $\{U_i\}$  are obtained through a polar decomposition of  $E_i$  and, following the Knill–Laflamme condition, are mutually orthogonal  $P_C U_j^\dagger U_i P_C = \delta_{ij} P_C \forall i, j$  [52]. In the measurement-feedback QEC paradigm, a correction protocol consists in coupling the system to an auxiliary one which after being carefully measured projects the state into one of the error subspaces. The measurement information allows then to bring the system's state back into the codespace through  $U_i^\dagger$ . In the dissipative QEC setting, these processes are performed by the incoherent Liouvillian  $\mathcal{L}_c$ .

### Lookup-table method and its limitations

A naive solution for the correction jump operators is to

consider a set  $\{L_{c,j}\}$  such that

$$\mathcal{L}_c(\rho) = \Gamma_c(\mathcal{R}(\rho) - \rho) \quad (4)$$

with  $\mathcal{R}$  being the recovery map in Eq. (3) and  $\Gamma_c$  the correction rate. This generates a time evolution described by  $\rho(t) = e^{-\Gamma_c t} \rho(0) + (1 - e^{-\Gamma_c t}) \mathcal{R}(\rho(0))$  which ensures that the system converges to the codespace. From the construction of the correction map presented above and assuming that  $P_E \equiv 0$ , we can see that the jump operators

$$L_{c,i} = \sqrt{\Gamma_c} U_i^\dagger P_i \quad (5)$$

realize the desired Liouvillian. If  $P_E \neq 0$ , additional jump operators are necessary to prevent the system to get stuck in the uncorrected subspace [24, 45]. However, as noted by Pastawski *et al.* [39] and more recently by Shtanko *et al.* [49], this solution is suboptimal for several reasons. To illustrate this, let us provide a concrete example.

One of the most studied dissipative QEC codes is the three-qubit bit-flip code [20–22, 41, 44, 45]. In this setting, a single logical qubit is encoded in a system of three qubits using  $\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}$ . It is protected against single qubit bit flips  $\{L_{e,i} = \sqrt{\Gamma_e} X_i\}_{i=1}^3$ , i.e. single-qubit Pauli  $X$  errors at a rate  $\Gamma_e$ , using

$$L_{c,i} = \sqrt{\Gamma_c} X_i \frac{1 - Z_i Z_j}{2} \frac{1 - Z_i Z_l}{2} \quad (6)$$

where  $i, j, l \in \{1, 2, 3\}$ . This operator effectively flips the  $i$ th qubit at a rate  $\Gamma_c$  if and only if its parity disagrees with the one of the two other qubits. In this example,  $X_i$  corresponds to the unitary  $U_i^\dagger$  from Eq. (5) while the projector  $P_i$  is built using operators  $Z_i Z_j$  and  $Z_i Z_l$  querying the parity of pairs of qubits. This construction can be straightforwardly generalized to other stabilizer codes [23, 43]. Every  $[[n, k, d]]$  code encoding  $k$  logical qubits in an  $n$ -qubit system and robust up to  $d$  errors lives in a  $2^k$ -dimensional subspace of  $\mathcal{H}$  that is invariant under the group generated by  $n - k$  independent operators  $S_i$ . Using these so-called generators, we can construct the set of projectors  $\{P_i\}_{i=1}^{N_{\text{sub}}}$  onto  $N_{\text{sub}} = 2^{n-k} - 1$  nontrivial and orthogonal error subspaces,

$$P_i = P(\vec{s}_i) := \prod_{j=1}^{n-k} \frac{1}{2} \left( 1 + (-1)^{\vec{s}_i[j]} S_j \right), \quad (7)$$

where  $\vec{s}_i \in \{0, 1\}^{n-k}$  and  $\vec{s}_i[j]$  the  $j$ -th element. The set  $\{L_{c,i}\}$  is then obtained using Eq. (5) resulting in  $N_{\text{sub}} = \mathcal{O}(2^n)$  jump operators. For non-stabilizer codes, a similar approach can be employed by introducing additional preventive jump operators that map each state in the orthonormal basis spanning the uncorrectable subspace into the correctable subspace [24, 45]. However, in this case, the number of required jump operators also scales exponentially with the system size.

This highlights a key limitation of the standard dissipative QEC framework. While the general construction in Eq. (5) allows the dissipative correction of  $\mathcal{C}$ , it comes at a significant operational cost: the number of required jump operators scales as  $\mathcal{O}(2^n)$ , where  $n$  is the number of qubits encoding the quantum information. Since each  $L_{c,i}$  can be implemented using  $\mathcal{O}(n)$  auxiliary qubits, this results in a tradeoff between space and time complexity. Specifically, one either requires an exponentially large auxiliary system with corrective jumps executed in parallel, or an exponentially long time to implement the correction map  $\mathcal{R}$  by recycling the auxiliary qubits. This limitation is also observed in measurement-feedback quantum error correction. Indeed, what we constructed here is a lookup-table type decoder which is known to be highly inefficient because of the exponentially growing table with the code size (we will discuss approximate decoders later on). All this naturally raises a critical question: Can dissipative QEC be achieved using only a polynomial number of jump operators, and what are the necessary conditions for this to be feasible?

Before we answer these questions, we must highlight a second, more fundamental limitation of the Liouvillian in Eq. (4). Let us consider the quantum trajectories picture of the dynamics [53, 54] which is equivalent to the Lindblad equation [55]. In this setting, the system is subject to a coherent evolution given by the so-called non-Hermitian Hamiltonian  $H_{\text{NH}} = -i/2 \sum L_j^\dagger L_j$  that is interleaved with incoherent jumps  $L_j$ . The dynamics is discretized in  $\delta t$  steps after which a quantum state  $|\psi(t)\rangle$  is updated to a non-normalized state  $|\psi(t + \delta t)\rangle$  whose norm is reduced  $\delta p = i \delta t \langle \psi(t) | H_{\text{NH}} - H_{\text{NH}}^\dagger | \psi(t) \rangle$ . The occurrence of a quantum jump follows a Bernoulli process with a probability  $\delta p$  where each  $L_j$  occurs with a probability  $\delta p_j / \delta p$  and  $\delta p_j = \delta t \langle \psi(t) | L_j^\dagger L_j | \psi(t) \rangle$ . Let us now consider the simple situation of a system subjected to unitary error operators  $L_{e,i}$  each occurring at a rate  $\Gamma_e$  as well as to a dissipative correction with jumps given in Eq. (5). Denoting  $N_e$  the number of different errors, the probability of any jump to occur at a time  $t$  equals

$$\delta p = \delta t (N_e \Gamma_e + \Gamma_c) \quad (8)$$

among which the probabilities that this jump corresponds to an error and correction are

$$p_e = \frac{N_e \Gamma_e}{N_e \Gamma_e + \Gamma_c} \quad \text{and} \quad p_c = \frac{\Gamma_c}{N_e \Gamma_e + \Gamma_c}. \quad (9)$$

This illustrates the second limitation of established dissipative QEC models: if the number of noise processes  $L_{e,i}$  increases polynomially with the system size, the correction rate must scale as  $\mathcal{O}(\text{poly}(n))$  to maintain a constant ratio  $p_e/p_c$ . In other words, for a given error rate and correction rate, the time between recovery operations and, consequently, the logical error rate, remain constant and independent of the code  $C$ . This observation has been noted in Refs. [39, 49]. In the latter work,

the authors derived explicit bounds on the logical error rate and concluded that dissipative qubit codes, employing a Liouvillian of the form given in Eq. (4), can achieve faster-than-polynomial reduction of the logical error rate when  $\Gamma_c \sim \mathcal{O}(\log(n))$ .

#### Issues with alternative solutions

An alternative solution to this problem is to consider dissipative QEC processes described by a more general Liouvillian  $\mathcal{L}_c$  than the construction in Eq. (4). Indeed, although the time evolution generated by Eq. (4) results in a steady state that lives in the logical subspace, at intermediate times the system's state remains mainly unchanged. Quantitatively, the trace distance between  $\rho(t)$  and the steady state  $\mathcal{R}(\rho(0))$  evolves independently of the code  $\mathcal{C}$ , i.e.  $\|\rho(t) - \mathcal{R}(\rho(0))\|_1 = e^{-\Gamma_c t} \|\rho(0) - \mathcal{R}(\rho(0))\|_1$ . In order to have a super-exponential convergence in the distance, a Liouvillian should instead generate a time evolution of the form

$$\rho(t) = e^{-\Gamma_c t} \mathcal{K}_t(\rho(0)) + (1 - e^{-\Gamma_c t}) \mathcal{R}(\rho(0)) \quad (10)$$

with  $\mathcal{K}_t$  being a time-dependent CPTP map with a nontrivial action on the error subspaces. When  $\|\mathcal{K}_t(\rho(0)) - \mathcal{R}(\rho(0))\|_1 < \|\mathcal{K}_{t'}(\rho(0)) - \mathcal{R}(\rho(0))\|_1$  for  $t < t'$ , we can think of it as a partial recovery map [49] which reduces the error weight without fully recovering the codespace. Pastawski *et al.* [39] demonstrated that such Liouvillians exist by proposing a construction where the  $\mathcal{L}_c$  is a sum of recovery operations acting on different stages of a concatenated QEC code whose strength decay as a power law with the level concatenation, i.e.

$$\mathcal{L}_c(\rho) = \sum_i \Gamma_c^{(i)} (\mathcal{R}_i(\rho) - \rho) \quad (11)$$

where  $\Gamma_c^{(i)} \sim \mathcal{O}(\Gamma^i)$  is the correction rate at the concatenation level  $i$  ( $i = 1$  being the physical level) and  $\Gamma < 1$  – a constant. Although Ref. [39] showed that partial recovery maps for dissipative QEC can be constructed when one considers specific concatenation of  $[[n, 3, 1]]$  codes, it remains unclear how to design such Liouvillians for general codes, how they perform against the lookup-table solution in Eq. (4), and, crucially, how many corrective jump operators must be engineered and what are the necessary resources.

#### The trickle-down solution

Devising a correction whose dynamics follows Eq. (10) is a non-trivial task that requires a careful design of the jump operators. The starting point of our derivation is to note that, although the number of error subspace  $N_{\text{sub}}$  scales exponentially with the system size, the number of jump operators  $L_{e,i}$  often increases only polynomially with  $n$ . As mentioned earlier, this difference comes from the fact that the error operators  $E_i$  are arbitrary products of the jump operators drawn from  $\mathbb{L}$ . We can thus partition  $\mathbb{E}$  into  $i$ -th order error subsets  $\mathbb{E}_i$  such

that  $\mathbb{E}_0 = \{\mathbb{1}\}$ ,  $\mathbb{E}_1 = \mathbb{L}$ ,  $\mathbb{E}_2 = \{O_i O_j \mid O_k \in \mathbb{L}, i \neq j\}$ , and  $\mathbb{E} = \bigcup_i \mathbb{E}_i$ . We can proceed identically with the error subspaces and their projectors, i.e.  $\mathbb{P}_0 = \{P_{\mathcal{C}}\}$ ,  $\mathbb{P}_1 = \{P_i \text{ associated to } E_i \in \mathbb{E}_1\}$ , etc. The key observation is that similarly to the Knill–Laflamme condition which sets the relationship between the logical and error subspaces we can derive a condition on the relationship between the error subspaces themselves:

Consider a  $q$ th-order error subspace with a projector  $P_i^{(q)} \in \mathbb{P}_q$  associated to the error  $E_i^{(q)} \in \mathbb{E}_q$ . This subspace is transformed to a  $(q+p)$ th-order error subspace under the action of  $p$ th-order errors. If the chosen code  $\mathcal{C}$  is capable of correcting errors up to order  $\ell \geq q+p$ , we get the following condition

$$\begin{aligned} & P_i^{(q)} E_j^{(p)\dagger} E_k^{(p)} P_i^{(q)} \\ & \propto U_i^{(q)} \left[ P_{\mathcal{C}} E_i^{(q)\dagger} E_j^{(p)\dagger} E_k^{(p)} E_i^{(q)} P_{\mathcal{C}} \right] U_i^{(q)\dagger} \\ & \propto U_i^{(q)} \delta_{jk} P_{\mathcal{C}} U_i^{(q)\dagger} = \delta_{jk} P_i^{(q)}. \end{aligned} \quad (12)$$

where we use the definition of the projectors  $P_i$ , the recovery unitary  $U_i^{(q)}$  from the polar decomposition of  $E_i^{(q)}$ , and the Knill–Laflamme condition for the  $(q+p)$ th-order errors. This implies that the codespace  $\mathcal{C}$  and the subspace  $\mathbb{P}_i^{(q)}$  subjected to  $E_j^{(p)}$  can both be recovered by the same unitary  $U_j^{(p)\dagger}$  obtained from the polar decomposition of this error operator. In other words, a jump operator of the form

$$L_{c,i}^{(p)} = \sqrt{\Gamma_c} U_i^{(p)\dagger} \left( P_i^{(p)} + P_i^{(q+p)} \right) \quad (13)$$

would project error states from  $\mathbb{P}_i^{(p)}$  back into the codespace  $\mathcal{C}$  while at the same time reducing the order of some error states from  $q+p$  to  $p$ . In this manner, we can design a set of jump operators that correct errors in a trickle-down fashion where every jump operator induces a complete recovery together with a reduction of the error weight. It is straightforward to see that the lowest number of jump operators would here be limited by the cardinality of  $\mathbb{L}$ . This minimal set would then be composed of

$$L_{c,i}^{(1)} = \sqrt{\Gamma_c} U_i^{(1)\dagger} \sum_{j=1}^{\ell} \sum_{k=1}^{K_j} P_{i,k}^{(j)} \quad \forall i \in \{1, \dots, |\mathbb{L}|\} \quad (14)$$

where  $U_i^{(1)\dagger}$  are the recovery unitaries associated to the error  $E_i^{(1)} \in \mathbb{E}_1 \equiv \mathbb{L}$  and  $P_{i,k}^{(j)} \in \mathbb{P}_j$  are the projectors onto  $j$ th-order error subspaces  $\mathbb{P}_i^{(j)}$  related to a  $(j-1)$ th-order subspaces through the error process  $E_i^{(1)}$ , i.e.  $P_{i,k}^{(j)} = U_i^{(1)} P_k^{(j-1)} U_i^{(1)\dagger}$  for some index  $k$ . This index runs from 1 to  $K_j = |\mathbb{P}_{j-1}| - 1$  where the  $-1$  comes from the fact that we must exclude from the set  $\mathbb{P}_{j-1}$  the projector onto the subspace  $\mathbb{P}_i^{(j-1)}$  related to  $(j-2)$ th-order

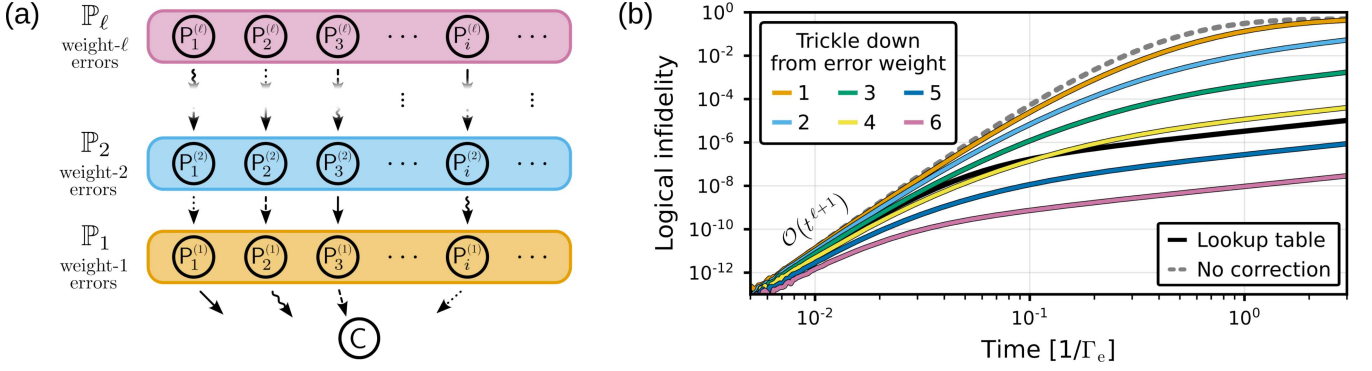


FIG. 2: *Trickle-down QEC.* (a) Schematic of the trickle-down dissipative error correction scheme. Error subspaces  $\mathbb{P}_i^{(j)}$  are grouped by their error weight  $j$ , with each horizontal block  $\mathbb{P}_j$  representing all subspaces at weight  $j$ . The number of subspaces per level increases with the number of fundamental noise processes. Error weights range up to  $\ell$ , the maximum correctable weight. Arrows indicate the action of trickle-down jump operators, where identical arrow styles denote transitions implemented by the same operator. The Knill-Laflamme condition in Eq. (12) determines which subspace pairs can be corrected by a common operator. (b) Logical infidelity over time for a 13-qubit repetition code initialized in  $|\psi_L\rangle \propto |0_L\rangle + i|1_L\rangle$ , subject to bit-flip noise (with error rate  $\Gamma_e = 10^{-2}\Gamma_c$ ) and dissipative QEC. We compare the performance of the lookup-table approach and the trickle-down scheme, the latter is performed up to different error-weight levels.

subspaces through the error process  $E_i^{(1)}$ . Given the number  $|\mathbb{L}|$  of weight-1 error operators, the total number of projectors  $P_{i,k}^{(j)}$  in each jump operator  $i$  is a simple combinatorial

$$N_{\text{proj}} = \sum_{j=1}^{\ell} \binom{|\mathbb{L}| - 1}{j - 1}. \quad (15)$$

We can now verify that our construction in Eq. (14) allows for a better ratio of the error and correction probabilities  $p_e/p_c$  (see Eqs. (9)) than using the naive jump operators in Eq. (5). In our case, we get

$$\frac{p_e}{p_c} = \frac{N_e \Gamma_e}{N_{\text{proj}} \Gamma_c}. \quad (16)$$

We see that the ratio is dependent on the chosen code  $C$  which sets the order  $\ell$  of the largest correctable errors and thus the total number of projectors  $N_{\text{proj}}$  which scales exponentially with the ratio  $\ell/|\mathbb{L}|$  (see Methods). The choice of the code also determines the number of jump operators that must be engineered. Indeed, considering a system with the only error processes being single qubit jump operators acting uniformly on every qubit, two codes  $C_1$  and  $C_2$  both correcting errors up to the order  $\ell$  but involving different number of physical qubits (e.g., Shor and Steane codes) would require a different number of jump operators.

#### Repetition code

Let us illustrate these results on a toy example, the repetition code. This code is a simple  $[[n, 1, n]]$  code encoding a single logical qubit in  $n$  physical ones. It is protected against single qubit bit-flip errors  $X_i$  at a rate  $\Gamma_e$  up to

order  $\ell = (n - 1)/2$ . The code subspace  $C$  is spanned by the states  $\{|0_L\rangle = |0\rangle^{\otimes n}, |1_L\rangle = |1\rangle^{\otimes n}\}$ . The projectors onto the error subspaces can be constructed using Eq. (7) and the generators of the code's stabilizer group,  $\{Z_1 Z_2, Z_2 Z_3, \dots, Z_{n-1} Z_n\}$ . Alternatively, we can define these projectors using  $P_C$  and the error operators. For example, for single qubit bit-flip subspaces the  $n$  projectors are given by  $P_i = X_i P_C X_i \in \mathbb{P}_1$ . The projectors into higher order error subspaces are obtained in a similar way. This construction is more convenient for our need as it allows us to build the desired jump operators using Eq. (14), which we can also rewrite as

$$L_{c,i}^{(1)} = \sqrt{\Gamma_c} \sum_{j=1}^{\ell} \sum_{k=1}^{K_j} P_{i,k}^{(j-1)} X_i \quad \forall i \in \{1, \dots, n\} \quad (17)$$

using the fact that the recovery unitaries  $U_i^{(1)\dagger}$  are exactly the bit-flip operators  $X_i$  thanks to their involutory character,  $X_i^2 = \mathbb{1}$ . The system dynamics in Eq. (1) is then governed by  $n$  jump operators competing against  $n$  error processes. Disregarding their possibly very nonlocal aspect (necessitating all-to-all connectivity), engineering  $n$  jump operators represents a much more tractable exercise than realizing the lookup-table solution which requires  $2^{n-1} - 1$  correction operators.

The two solutions also differ in error correction performances. We study the performance of the jump operators in Eqs. (5) and (17) by simulating the dynamics of a 11-qubit repetition code initialized in the state  $|\psi_L\rangle \propto |0_L\rangle + i|1_L\rangle$  and subjected to bit-flip error  $L_{e,i} = \sqrt{\Gamma_e} X_i$  and correction jumps. After an evolution time  $t < 3/\Gamma_e$  we decode the system state by majority voting and compute the logical fidelity with the initial

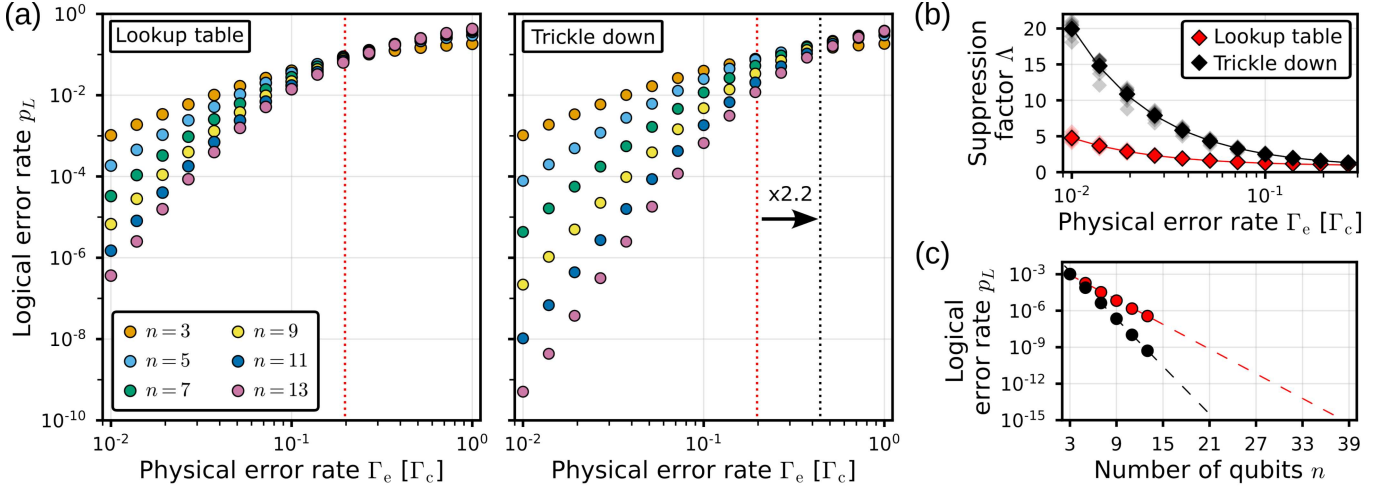


FIG. 3: *Trickle-down vs. lookup-table QEC.* (a) Logical error rate  $p_L$  as a function of the physical error rate  $\Gamma_e$  for lookup-table (left) and trickle-down (right) dissipative correction of  $n$ -qubit repetition codes under bit-flip noise. Each data point is extracted from fits to time evolutions shown in Fig. 2(b) (see main text). Dotted vertical lines indicate the threshold value below which increasing the code size  $n$  results in exponential suppression of  $p_L$ . The thresholds are  $\sim 0.2$  and  $\sim 0.44$  for both solutions, respectively. (b) Exponential suppression factor  $\Lambda = \Gamma_e/\Gamma_e^*$  as a function of code size  $n$  for the lookup-table (red) and trickle-down (black) approaches. Data points correspond to  $(p_{L,i}(\Gamma_e)/p_{L,i+k}(\Gamma_e))^{1/k}$ , with  $p_{L,i}(\Gamma_e)$  the logical error rate for an  $i$ -qubit code (excluding  $i = 3$ ) at fixed  $\Gamma_e$ . Shaded markers show individual data points; solid markers show their averages. Solid lines are fits to inverse scaling, yielding  $\Gamma_e^* \approx 0.04$  and  $\Gamma_e^* \approx 0.2$  for lookup-table and trickle-down schemes, respectively. (c) Number of qubits required to achieve a target logical error rate of  $10^{-15}$  at a fixed physical error rate  $\Gamma_e = 10^{-2}\Gamma_c$ . Dashed lines show extrapolations based on the extracted  $\Gamma_e^*$ .

state. The results are presented in Fig. 2(b). In addition, to simulate the optimal correction operators, we also perform simulations with jump operators correcting errors only up to orders  $m < \ell = 5$  (i.e. we build the jump operators in Eq. (17) with the first sum going only to  $m$ ). At short timescales all infidelity curves follow an identical trend dictated by the final decoding step of the simulation using majority vote. This decay of the fidelity described by  $\mathcal{O}(t^{\ell+1})$  comes from the unavoidable probability that multiple errors arise and accumulate resulting in an error of weight at least  $\ell + 1$  and a misdecoding of the system state. Thus, majority vote alone allows a short timescales to slow down the loss of information about the initial logical state by using codes with larger  $\ell$ . The goal of an error correction protocol is however to (1) reduce the logical error rate at longer timescales, and (2) start this secondary regime at the earliest time possible. A clear example of this is the infidelity evolution using the lookup-table correction jump operators (in black in Fig. 2(b)). We observe that for  $t \gtrsim 0.05/\Gamma_e$  the logical infidelity starts to increase at a much slower rate than for earlier times such that at  $3/\Gamma_e$  the infidelity is only  $\sim 10^{-4}$  as opposed to 0.5 in the uncorrected case. We now compare this to the performance of our trickle-down solution. When considering correction jump operators with projectors up to error weight  $m < 4$ , the logical fidelity is lower than for the lookup-table solution. This

is expected as the error processes are not fully corrected resulting in always growing uncorrectable error subspace. However, when  $m = 4$  and  $m = \ell$ , our trickle-down solution outperforms the lookup-table one and, in the optimal case, by more than an order of magnitude. We also notice that for  $m = \ell$ , the onset of the slow increase of the logical infidelity is shifted to earlier timescales by a factor of 5 which is a signature that correcting errors in a trickle down fashion enhances the probability that a correction jump occurs. In other words, it enhances the ratio  $p_e/p_c$ .

Quantitatively, the long timescale trend of the infidelity curves can be described by the function  $F(t) = 0.5 [1 + \exp(\epsilon(t - \tau))]$  where  $\epsilon$  represents the rate of the decay and  $\tau$  the time at which one switches from one regime to another. The form of this function comes straightforwardly from a combinatorial counting of the total number of errors that occurred in the system (e.g. see derivation Ref. [56]). Given that ideally we would like both values to be low, their product is a good measure of the error correction performance. We can interpret the dimensionless quantity  $p_L = \epsilon \tau$  as the probability to misidentify the logical state. When  $\tau$  is given in units of  $1/\Gamma_e$  then  $p_L$  becomes the logical error rate commonly used in measurement based QEC where it represents the frequency with which the decoder fails. We evaluate the error correction performance of the two approaches by

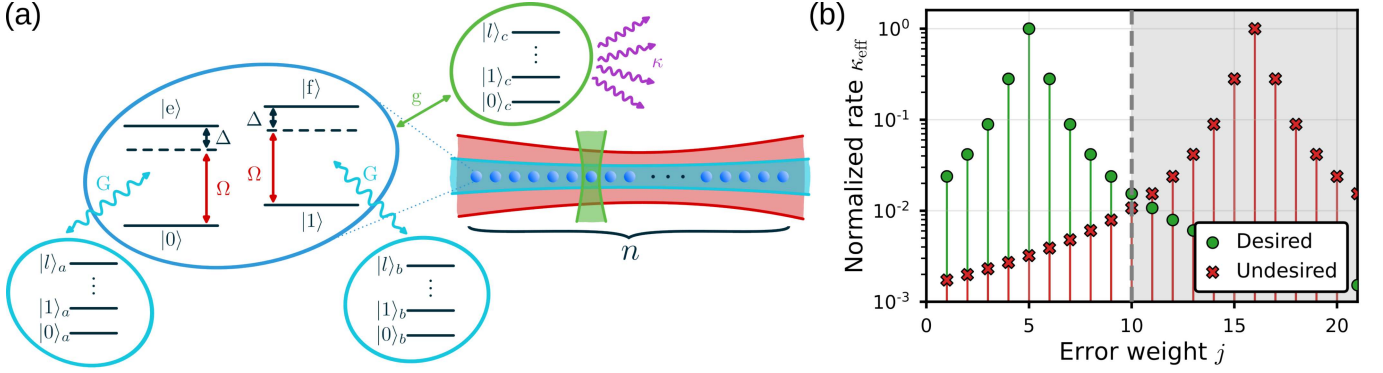


FIG. 4: *Trapped-ion implementation.* (a) Schematic of a trapped-ion setup for implementing trickle-down correction of a repetition code. Each of the  $n$  qubits is encoded in two electronic levels of a single ion. The correction mechanism uses two additional electronic levels  $|e\rangle$  and  $|f\rangle$ , two global motional modes  $a$  and  $b$ , and a local mode  $c$  that is sympathetically cooled to remove entropy. The process is driven by two global laser beams implementing the interactions in Eqs. (18) and (19), and local beams implementing Eq. (20). (b) Effective rates of the jump operators from Eq. (21) for a 21-qubit repetition code, shown as a function of the error weight of the target subspaces. Desired processes reduce the error weight via  $j \rightarrow j - 1$  transitions, while undesired ones increase it via  $j \rightarrow j + 1$ . Shaded regions correspond to processes beyond the correctable error weight  $\ell = 10$ , which are inactive but included to highlight the symmetry of the Lorentzian profiles.

fitting this function to the infidelity curves using the Levenberg–Marquardt algorithm that we bias using  $t^{\ell+1}$  to avoid overfitting the dynamics at short times [56]. We do this for various code sizes  $n \in \{3, 5, 7, 9, 11, 13\}$  and error rates  $\Gamma_e \in [10^{-2}, 10^0] \times \Gamma_c$ . We plot in Fig. 3(a) the logical error rate  $p_L$  as a function of  $\Gamma_e$  for the lookup-table correction jump operators in Eq. (5) and our trickle-down solution in Eq. (17). Both situations show that there exist a threshold below which increasing the code size, exponentially decreases the logical error rate. Yet, we observe that the threshold of the trickle-down solution is  $\sim 2.2$  larger than the lookup-table one (see details below). Moreover, in this sub-threshold regime, we observe that the trickle-down solution outperforms the lookup-table one for all  $n > 3$ . For  $n = 3$ , both solutions are equal. In this regime, the logical error rate follows the function  $p_L \propto (\Gamma_e/\Gamma_e^*)^{\ell+1}$  with  $\Gamma_e^*$  a constant related to the threshold value. A figure of merit is the exponential error suppression factor  $\Lambda = \Gamma_e^*/\Gamma_e$  which can be obtained from the ratio of  $p_L$  for different  $n$ . Specifically, we calculate  $(p_{L,i}(\Gamma_e)/p_{L,i+k}(\Gamma_e))^{1/k}$  for different  $k$  with  $p_{L,i}(\Gamma_e)$  being the logical error rate for an  $i$ -qubit code and an error rate  $\Gamma_e$ . These quantities are then averaged out to estimate the exponential suppression factor for a given  $\Gamma_e$ . We plot  $\Lambda$  as a function of the error rate in Fig. 3(b) for both solutions and show that the trickle down one allows to increase  $\Lambda$  by a factor of  $\sim 4$ . From these results, we obtain an estimate of  $\Gamma_e^*$  by fitting an inverse proportionality function  $f(\Gamma_e) = A/\Gamma_e + B$  to the data (the offset  $B$  is here to capture the behavior of  $\Lambda$  in the high error rate regime). We obtain  $\Gamma_e^* = 0.04$  for the lookup table solution and  $\Gamma_e^* = 0.2$  for the trickle down one. In addition, we find that the positions of the thresh-

olds (dotted lines in Fig. 3(a)) are located coincidentally at the square root of  $\Gamma_e^*$  for both solutions with values  $\sim 0.2$  and  $\sim 0.44$ , respectively. As a final performance metric, we use  $\Gamma_e^*$  to extrapolate the number of qubits  $n$  necessary to achieve a logical error rate  $p_L = 10^{-15}$  given a physical error rate  $\Gamma_e = 10^{-2}$ . The results are shown in Fig. 3(c) where the dashed lines follow  $p_L \propto (\Gamma_e/\Gamma_e^*)^{\ell+1}$  with  $\ell = (n+1)/2$  and proportionality constants 0.02 and 0.7 for the lookup-table and trickle-down solutions, respectively. The plot shows that performing error correction in a trickle-down fashion requires  $\sim 21$  qubits as opposed to  $\sim 37$  when using the standard approach.

Interestingly, the estimate for the lookup table solution aligns with the most recent experimental value of  $\Lambda$  for a measurement-feedback repetition code relying on superconducting qubits [7]. In this work, the Google Quantum AI team obtained an exponential suppression factor of  $8.4 \pm 0.1$  for an averaged physical error probability slightly below  $10^{-2}$ . These values are comparable with our results in Fig. 3 when extrapolated to smaller  $\Gamma_e$  values. This is a promising result as it suggests that the trickle-down solution if implemented in a realistic experiment could outperform the established error correction protocols by several orders of magnitude.

#### Trapped-ion implementation

We now present a potential implementation of such a repetition code corrected using the trickle-down solution in a trapped-ion platform. The scheme we propose corresponds to an extension of the one presented in Ref. [44]. Let us consider a system of  $n$  ions each possessing four addressable internal energy levels,  $\{|0\rangle, |1\rangle, |e\rangle, |f\rangle\}$ , and three bosonic modes  $a$ ,  $b$  and  $c$  corresponding to motional

modes of the ion string. We utilize these additional levels and modes to engineer the correction jump operators acting onto the computational subspace spanned by the state  $\{|0\rangle, |1\rangle\}^{\otimes n}$ . The engineering is performed through reservoir engineering techniques, i.e. the coupling of the system using a Hamiltonian  $H$  and a tunable nonunitary (i.e. entropy removing) dissipation channel such as optical pumping or sympathetic cooling [57]. In our case illustrated schematically in Fig. 4(a),  $H$  consists of three parts: First, a global laser drive addressing simultaneously  $|0\rangle \leftrightarrow |e\rangle$  and  $|1\rangle \leftrightarrow |f\rangle$  with a strength  $\Omega$  and detuned by  $\Delta$  from the resonance

$$H_1 = \sum_{i=1}^n \Delta (|e\rangle\langle e|_i + |f\rangle\langle f|_i) + \frac{\Omega}{2} (|e\rangle\langle 0|_i + |f\rangle\langle 1|_i) + \text{H.c.} \quad (18)$$

which we can also split into the sum of a rest  $H_0$  and perturbation  $V$  parts. Next, we require a second global laser beam now coupling  $|0\rangle / |f\rangle$  and  $|1\rangle / |e\rangle$  to the motional modes  $a$  and  $b$ , respectively, with a strength  $G$  and a detuning  $\delta$ ,

$$H_2 = \delta (a^\dagger a + b^\dagger b) + G \sum_{i=1}^n a^\dagger |1\rangle\langle e|_i + b^\dagger |0\rangle\langle f|_i + \text{H.c.} \quad (19)$$

Third, we add a single addressing laser coupling an ion at position  $i$  to an auxiliary motional mode  $c_i$

$$H_{3,i} = g c_i^\dagger (|0\rangle\langle e|_i + |1\rangle\langle f|_i) + \text{H.c.} \quad (20)$$

We use this mode to remove entropy from the system through jump operators  $L_i = \sqrt{\kappa} c_i$  with a tunable rate  $\kappa$ . The error correction mechanism behind the total Hamiltonian  $H = H_1 + H_2 + H_3$  is the following: using  $H_2$  in a strong-driving regime creates many-body resonances between computational states and superpositions of states with a single excitation  $|e\rangle$  or  $|f\rangle$ . Driving  $H_1$  in a perturbative fashion and carefully setting both detunings  $\Delta$  and  $\delta$ , we can address specific transitions between those states. By adiabatically eliminating the local mode  $c_i$  and excited states  $|e\rangle$  and  $|f\rangle$ , we can show that the effective dynamics of the system is dissipative with a jump operator (see Supplemental Material)

$$L_{\text{eff},i} = \sum_{j=1}^{\ell} \sqrt{\kappa_{\text{eff}}(j)} (|0\rangle\langle 1|_i P_1^{(j)} + |1\rangle\langle 0|_i P_0^{(j)}) + \sqrt{\kappa_{\text{eff}}(n-j)} (|1\rangle\langle 0|_i P_1^{(j)} + |0\rangle\langle 1|_i P_0^{(j)}) \quad (21)$$

where  $P_\psi^{(j)}$  indicate projectors onto all computational state that are  $j$  bit flips away from the logical state  $|\psi_L\rangle$ . The first parenthesis in  $L_{\text{eff},i}$  displays the *desired* error correction operators, while the second one bares the *undesired* processes increasing the error weight instead of decreasing it. The rate of these two processes follow a

Lorentzian function (see Supplemental Material)

$$\kappa_{\text{eff}}(x) = \frac{\Omega^2}{\kappa_{\text{eng}}} \left[ \left( \frac{2G}{\kappa_{\text{eng}}\delta} \right)^2 \left( x - \frac{\delta\Delta}{G} \right)^2 + 1 \right]^{-1} \quad (22)$$

with  $\kappa_{\text{eng}} \propto g^2/\kappa$  being the rate of the engineered dissipation. Choosing the median of the Lorentzian  $x_0 = \delta\Delta/G$  to be an integer, the process correcting all error states of weight  $x_0$  and erroneous qubit  $i$  will be on resonance and happening at a rate  $\Omega^2/\kappa_{\text{eng}}$ . All the other processes will be off resonant. Importantly, the undesired ones will always stay off resonant as long as  $x_0 \in \{1, \dots, \ell\}$  since they follow  $\kappa_{\text{eff}}(n-j)$ , i.e. the same Lorentzian as for the desired processes, mirrored symmetrically around  $\ell$ . The normalized rates are represented in Fig. 4(b) for a system of  $n = 21$  qubits,  $x_0 = 5$ ,  $\Delta = 4\kappa_{\text{eng}}$ ,  $\delta = 3.46\kappa_{\text{eng}}$  and  $G = 1.66\kappa_{\text{eng}}$ . The ideal rate function would be a step function with a value of 1 for the desired processes and 0 for the undesired ones. It can potentially be achieved using inhomogeneous and amplitude modulated global drives rendering the parameters ion dependent. We leave this for future studies.

The desired process in Eq. (21) looks slightly different than the trickle down solution we proposed in Eqs. (14) and (17). However, they can be shown to be equivalent. It is straightforward to check that the projector  $P_0^{(j)}$  is related to the projectors  $P_{i,k}^{(j)}$  in Eq. (14) through the relation

$$|1\rangle\langle 0|_i P_0^{(j)} = |1\rangle\langle 1|_i \sum_{k=1}^{K_j} P_{i,k}^{(j)}. \quad (23)$$

The relation for  $P_1^{(j)}$  reads identically with the roles of  $|0\rangle$  and  $|1\rangle$  interchanged. Given that  $X_i = |1\rangle\langle 1|_i + |0\rangle\langle 0|_i$ , the sum of these expressions leads to the intended result. In other words, the desired part of the jump operator in Eq. (21) realizes to the proposed trickle down solution with the exception that the rates are function of the error weight  $j$ . To remedy this we can make the global drive  $H_2$  multi-tone. Carefully setting the strength  $G_x$  and detuning  $\delta_x$  of each tone puts several transitions on resonance (according to Eq. (22)). The optimal number of tones is  $\ell$  allowing to address all the desired processes at once. Finally, adding  $n$  single-ion addressing laser beams with  $n$  auxiliary motional mode  $c_i$ , i.e. a Hamiltonian  $H_{3,i}$  in Eq. (20) for every ion, engineers a dissipative process with  $n$  jump operators  $\{L_{\text{eff},i}\}_{i=1}^n$ .

In summary, our trapped-ion scheme realizes exactly the trickle down error correction of a repetition code using two global laser beams (among which an  $\ell$ -tone one) and  $n$  single-ion addressing ones. Such a setup would produce a noise biased quantum code performing as illustrated in Fig. 3. Moreover, the presence of undesired processes in Eq. (22) does not reduce this performance as their rates are sufficiently low. As a matter

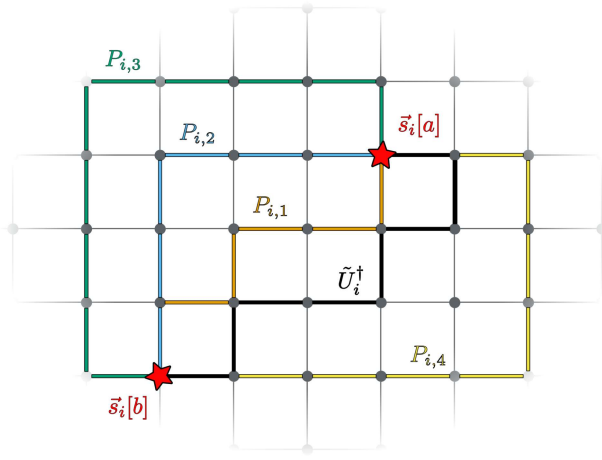


FIG. 5: *Approximate decoders.* Example of a decoder graph for a surface code. Nodes represent syndrome bits  $\vec{s}_i$ , and edges represent possible error paths (or equivalently, corrections). The decoder's goal is to connect each pair of flagged vertices (i.e.,  $\vec{s}_i = 1$ , marked with stars) via paths of minimal total weight, ideally matching the actual error. The original error is omitted. The decoder outputs a correction path  $\tilde{U}_i^\dagger$  for each pair. Due to the topological nature of surface codes, any path differing by a stabilizer (e.g.,  $P_{i,1}$ ,  $P_{i,2}$ ,  $P_{i,3}$ ,  $P_{i,4}$ ) is also a valid correction, as such loops are undetectable via stabilizer measurements and do not affect the logical state.

of fact, the trickle down simulation data presented in Fig. 3 were obtained by simulating the system dynamics under the multi-tone correction operators  $\{L_{\text{eff},i}\}_{i=1}^n$  with parameters set to experimentally achievable values ( $\delta = 10\kappa_{\text{eng}}$ ,  $\Omega = 3\kappa_{\text{eng}}$ ,  $\kappa_{\text{eng}} = 10$  kHz). The dynamics under the ideal jump operators given in Eq. (17) with rate  $\Gamma_c = \Omega^2/\kappa_{\text{eng}}$  leads to identical results. To reduce the experimental overhead of using  $n$  single-ion beams, the correction can be performed in a sequential manner by individually addressing subsets of ions one after the other. This would increase the logical error rate by a constant factor but its scaling with the number of qubits would remain unchanged compared to the one presented in Fig. 3.

#### Approximate decoders

Before concluding, we must put our trickle down solution within the context of the current most common ways of performing quantum error correction. As noted earlier, a lookup table with exponentially increasing size is unfeasible even in measurement-feedback QEC. Instead, this paradigm relies on approximate decoding algorithms, which use measurement outcomes to estimate the most likely error subspace into which the system has been projected. In the framework of stabilizer codes, these approximate decoders take as input the binary vector  $\vec{s}_i$  (the syndrome) obtained from stabilizer measurements

and output a correction unitary  $\tilde{U}_i^\dagger$  that ideally approximates the desired  $U_i^\dagger$  (recall Eqs. (3) and (7)). To further simplify the problem, decoders often incorporate a model of the error process in the system, for example, the locality of errors determined by qubit connectivity and single- and two-qubit error probabilities obtained from device calibration. For codes where each independent error produces a syndrome  $\vec{s}_i$  with at most two flagged bits  $\vec{s}_i[a] = 1 = \vec{s}_i[b]$  (as in surface codes), the estimation can be reformulated as a graph problem [58]. Here, the  $n - k$  nodes correspond to the entries of  $\vec{s}_i$  and the edges represent the errors (or, equivalently, the required corrections). We show an example of such a decoder graph in Fig. 5. The decoder's task is to match each pair of flagged vertices with a path (i.e. an error process) with the lowest possible total weight (with the path weight calculated efficiently using the available information). Various algorithms can address this problem, among which the minimum-weight perfect matching algorithm is the most common [59–62]. For further details about decoders, see Refs. [58, 63]. Translating approximate decoders from measurement-feedback to dissipative QEC, one may define the correction jump operators as

$$L_{c,i} = \sqrt{\Gamma_c} \tilde{U}_i^\dagger(\vec{s}_i) P_i \quad (24)$$

where  $\tilde{U}_i^\dagger(\vec{s}_i)$  is the recovery unitary produced by the approximate decoder for the syndrome  $\vec{s}_i$  and  $P_i$  is the projector onto the  $i$ th error subspace. These operators are an approximate version of the lookup-table operators in Eq. (5) with the advantage of limiting the exponential growth of the number of correction jumps with  $n$  by excluding the correction of the most unlikely error processes given the error model. However, they still suffer from the requirement  $\Gamma_c \sim \mathcal{O}(\text{poly}(n))$  (see Eq. (9)). To improve this aspect, one must consider the topological properties of certain codes. For example, in surface codes, a closed path in the decoder graph may go undetected because no syndrome bit  $\vec{s}_i[j]$  is flagged; consequently, such a path would not be recognized as an error. Therefore, to correct a pair of flagged vertices, multiple closing-loop correction paths may be constructed. Conversely, a given unitary  $\tilde{U}_i^\dagger(\vec{s}_i)$  can recover the logical information from several error subspaces simultaneously, with projectors  $\{P_{i,j}\}$  (see examples in Fig. 5). Thus, the action of approximate decoders in measurement-feedback QEC should be modeled in the dissipative framework using the correction jump operators

$$L_{c,i} = \sqrt{\Gamma_c} \tilde{U}_i^\dagger(\vec{s}_i) \sum_j P_{i,j}. \quad (25)$$

These operators are similar to those proposed in Eq. (14) with the difference that the  $P_{i,j}$  are not necessarily projectors onto subspaces corresponding to a specific error order. Nevertheless, they lead to improved correction performance, analogous to the trickle down solution. That being said, both sets of jump operators in

Eqs. (24) and (25) are challenging to implement in practice, as they require not only knowledge of the syndrome  $\vec{s}_i$  but also the ability to engineer dynamically changing recovery unitaries  $\tilde{U}_i^\dagger(\vec{s}_i)$ . Nonetheless, this discussion highlights two key points: (1) approximate decoders overcome the issue of an exponentially increasing number of correction operators as the code size grows; and (2) a code's topology can induce effects similar to a trickle down correction. These observations also prompt questions regarding measurement-feedback QEC: Is there a sequence of stabilizer measurements whose outcomes can indicate which operation would most effectively reduce the error weight? Can one devise an approximate decoder that functions as a trickle down correction, and would this be more time-efficient? Recent works have begun to address the second question [64–66]. For instance, Ref. [64] demonstrated that combining outputs from multiple decoders and selecting the best local correction from each can improve both the error suppression factor  $\Lambda$  as well as the decoding time. This approach effectively selects the lowest-weight paths across decoder outputs and can be interpreted as minimizing the area of loops in the code (even though not considered as errors). Decoders proposed in Refs. [65, 66] follow a different strategy: they construct candidate error sequences by adding faults incrementally, guided by a decision tree, until the resulting syndrome matches the observed one. These algorithms could be adapted to prioritize corrections that reduce the overall error weight, rather than reconstructing the full error, thus bringing them closer in spirit to a trickle down correction strategy.

### Bosonic QEC

The trickle-down QEC is analogous to a cooling process, wherein the system is gradually driven towards its lowest energy configuration. In our context, the system is cooled into a degenerate subspace in a coherent way. The correction of bosonic codes is another type of QEC which is often framed as a type of cooling. This becomes evident through the subsystem decomposition approach, which partitions the infinite-dimensional Hilbert space of a quantum harmonic oscillator into a logical qubit subspace and an auxiliary gauge mode [67]. Rewriting the system dynamics in this form allows one to isolate and analyze the effects of both noise and correction on each subsystem.

A canonical example is the dissipative stabilization of the Cat qubit [26]. Consider a quantum harmonic oscillator with annihilation and creation operators  $a$  and  $a^\dagger$ . A dissipative dynamics governed by the jump operator  $L_{\text{cat}} = \sqrt{\Gamma_c}(a^2 - \alpha^2)$  drives the system into a degenerate ground state manifold spanned by the coherent states  $|\pm\alpha\rangle$ , which form the logical-Z basis of the Cat qubit [26, 68]. In the subsystem decomposition of this code (known as the shifted Fock basis) the logical states can be written as  $|\pm\alpha\rangle = |0/1\rangle \otimes |0\rangle$ , while for sufficiently

large  $\alpha$ , the jump operator becomes  $L_{\text{cat}} \propto I \otimes b$ , where  $I$  acts on the logical subspace and  $b$  lowers the gauge mode by one excitation [69]. Importantly, in this picture, noise such as dephasing also acts predominantly on the gauge mode, effectively exciting it. That is,  $a^\dagger a \approx I \otimes (b^\dagger b + |\alpha|^2 + ab + \alpha^* b^\dagger)$ . Thus, dissipative stabilization of the Cat qubit realizes a trickle-down correction of dephasing noise by gradually removing single excitations from the gauge mode, independent of the gauge mode's state. In contrast, a lookup-table correction approach would require jump operators that measure the gauge mode's excitation level and apply corrections accordingly. The number of correction jumps would then scale with the number of gauge mode excitations which one would like to correct.

Other bosonic codes also exhibit trickle-down correction. A notable example is the nonlinear generalization of Cat qubits realized through nonlinear reservoir engineering [70]. In their subsystem decomposition, the stabilization operator takes the form  $I \otimes f(b^\dagger b)b$ , where  $f$  is a nonlinear function of the gauge mode's excitation number. Similar to our trapped-ion scheme in Eq. (21), the function  $f$ , which is tunable via system parameters, controls the rate of trickle-down at different error weights, and has been shown useful to enhance robustness against noise [34, 70].

Our final examples are GKP codes. Designed to protect quantum information against small displacements in position and momentum, GKP codes use two dissipative jump operators—each targeting one quadrature [28, 71]. These operators induce a trickle-down correction process, as shown in Ref. [35], via a decomposition of the oscillator Hilbert space into error subspaces of increasing boson number gain or loss. Each jump operator acts to transfer the quantum state down this hierarchy, closer to the code subspace. The authors illustrate this process by showing trajectories of various error states through these subspaces during stabilization. The trickle down aspect of these jump operators can also be shown through their subsystem decomposition (known as the shifted grid or Zak basis) which parametrizes the gauge mode in terms of the modular version of position and momentum operators [67, 72]. Here, as in the Cat code example, the stabilization gradually displaces the gauge mode state towards its ground state.

In summary, many bosonic QEC schemes developed to date share a trickle-down character. This is largely due to the use of an auxiliary two-level system to mediate correction which intrinsically can only remove one bit of entropy at a time [73]. Trickling errors down represents a solution that optimizes the number of error states corrected simultaneously through one entropy removal round. Sivak *et al.* [35] qualified it as being an approach that reduces the control overhead in bosonic codes. In our work, we showed that this reduction is in fact exponential and that it is accompanied by an improvement

in the error correction performance. Thus, beyond the well-known benefit of reducing the number of physical systems required to encode a logical qubit, the rapid success of bosonic codes may also stem from the intrinsic efficiency of their QEC protocols. Nonetheless, the auxiliary qubit introduces its own challenges: errors in the qubit can propagate to the oscillator and cause logical failures, thereby undermining fault tolerance [18, 35, 71]. Solutions to this issue are a topic of ongoing research [74].

## Discussion

In this work, we introduced a novel construction of jump operators for dissipative quantum error correction protocols. Unlike the conventional lookup-table approach, which projects the system into a specific error subspace and applies a dedicated correction unitary, our method implements a trickle-down mechanism that reduces the system's error weight one level at a time. This construction scales efficiently with code size, reducing the number of required correction operators from exponential to polynomial. Beyond overhead reduction, the trickle-down approach improves correction performance, as demonstrated using the dissipative repetition code: at a physical error rate of  $10^{-2}$ , it achieves a fourfold increase in the exponential suppression factor  $\Lambda$  and lowers the logical error rate  $p_L$  by several orders of magnitude. This implies that reaching  $p_L = 10^{-15}$  requires only 23 qubits with trickle-down correction, compared to 37 with the lookup-table method. A naive extrapolation to surface codes suggests a qubit overhead reduction of  $1 - (23/37)^2 \approx 61\%$ .

From a practical perspective, we proposed an implementation of the dissipative repetition code in a trapped-ion platform. Our scheme uses global drives to induce coherent transitions between many-body error states and local beams for entropy removal, extending the scheme from Ref. [44]. Our framework connects autonomous QEC for discrete-variable codes with both measurement-feedback schemes using approximate decoders in topological codes and dissipative bosonic protocols mediated by auxiliary qubits. In this sense, our scalable construction, based on physically motivated principles akin to cooling, offers a unifying perspective with broad applicability across the field.

The main limitation of our method is its reliance on highly nonlocal operators, which poses challenges for hardware platforms constrained by local connectivity, such as superconducting qubit arrays. However, other leading platforms such as trapped ions and Rydberg atom arrays offer more flexibility in this regard (through a natural all-to-all connectivity or rearrangement of atoms). Moreover, current research in quantum error correction has seen a surge of interest in quantum low-density parity-check (qLDPC) codes [75, 76], whose stabilizers are low-rank but often nonlocal. In this context, nonlocality may not be a drawback, and it is worth

exploring whether our construction can further improve the logical error rates of qLDPC codes. Another interesting prospect is to learn as to whether measurement-feedback QEC can be implemented in a trickle-down fashion. Would this require measurements that differ from the stabilizer ones? If not, is there a classical decoder capable of outputting correction operations that reduce error weight and suitable for real-time decoding of quantum codes? Addressing these questions is left for future works.

Finally, our work focused on quantum codes functioning as quantum memories. Although these can be applied to quantum sensing and metrology [44], an intriguing direction for future research is to investigate the impact of trickle-down correction on logical quantum gates for quantum computation. Lebreuilly *et al.* [24] proved the existence of generalized error-transparent Hamiltonians, showing that if a physical operation transforms the codespace and all relevant error subspaces in the same way, then the correct logical gate is realized upon recovery. In other words, if the operation coherently increases the system's error weight (while retaining it below the uncorrectable level) then dissipative QEC can restore the system such that the overall process implements the intended logical gate. One example of this principle appears in error-corrected gates between GKP codes. These two-qubit gates defined for ideal infinite-energy codewords generate unwanted entanglement when applied to realistic finite-energy GKP states [25]. However, dissipative trickle-down QEC has shown to correct much of this spurious entanglement, thereby restoring the logical operation to high-fidelity [77]. An open question is how such error-transparent operations manifest in other codes and whether they could simplify the experimental requirements for implementing fault-tolerant logical gates. Beyond quantum error correction, our work connects to recent developments in dissipative quantum algorithms, where similar cooling schemes enable scalable protocols [78, 79].

## Materials and Methods

### Nomenclature

In the main text, we used the following nomenclature for the different mathematical fonts.

|                                           |                                                               |
|-------------------------------------------|---------------------------------------------------------------|
|                                           | Hilbert <u>spaces or subspaces</u>                            |
| H, C, P                                   | associated to codewords or to some projectors                 |
| $\mathcal{L}, \mathcal{R}, \mathcal{K}_t$ | <u>Superoperators</u> or <u>CPTP maps</u>                     |
| H, L, E, P, S, U                          | <u>Operators</u> representing Hamiltonians, jumps, etc.       |
| $\mathbb{L}, \mathbb{E}, \mathbb{P}$      | <u>Sets</u> of error operators, error subspaces or projectors |

### Number of projectors in each jump operator

The number of projectors in each trickle down jump operator follows a sum of binomial coefficients shown in Eq. (15). This accounts for the fact that a logical state can transition to an erroneous state of order  $m$  through any of the  $|\mathbb{L}|$  choose  $m$  possible paths. The jump operators in Eq. (14) then coherently reduce the error weight by one, driving all  $|\mathbb{L} - 1|$  choose  $m$  paths that achieve this. The number of projectors in each jump operator can then be lower bounded by assuming that the ratio  $\epsilon = \ell/|\mathbb{L}| < 1/2$  and using Stirlings formula. We get

$$\begin{aligned} N_{\text{proj}} &\geq \frac{2^{(|\mathbb{L}|-1)H(\epsilon)}}{\sqrt{8(\ell-1)\left(1 - \frac{\ell-1}{|\mathbb{L}|-1}\right)}} \\ &\geq \frac{2^{4(\ell-1)\left(1 - \frac{\ell-1}{|\mathbb{L}|-1}\right)}}{\sqrt{8(\ell-1)\left(1 - \frac{\ell-1}{|\mathbb{L}|-1}\right)}} \end{aligned} \quad (26)$$

where in the second inequality we are lower bounding the binary entropy function  $H(p) = -p \log_2(p) - (1-p) \log_2(1-p) \geq 4p(1-p)$ .

### Effective dynamics of the trapped-ion model

The trapped-ion implementation of the dissipative repetition code relies on the effective operator formalism [80] and resonance engineering [44] techniques. Before using them it is convenient to no longer work in the basis of individual internal states of ions but in a many-body basis involving in addition excitations in modes  $a$  and  $b$  introduced

$$|i_1, \dots, i_c\rangle = \left[ \left( \bigotimes_{k=1}^c |1\rangle\langle 0|_{i_k} \right) a^\dagger + \left( \bigotimes_{k=1}^c |0\rangle\langle 1|_{i_k} \right) b^\dagger \right] |\psi_L\rangle |0, 0\rangle \quad (27)$$

where  $|\psi_L\rangle = \alpha |0_L\rangle + \beta |1_L\rangle = \alpha |0\rangle^{\otimes n} + \beta |1\rangle^{\otimes n}$  is an arbitrary logical state and  $|0, 0\rangle$  represent the state of oscillators  $a$  and  $b$  in the Fock basis. These states represent the erroneous code states of weight  $c \leq \ell$  with exactly one excitation in either of the two modes. We

use  $\{i_1, \dots, i_c\}$  to indicate the indices of the corrupted physical qubits (we will refer to each individual ion in the state  $|0/1\rangle$  as physical qubits). The particularity of this definition is that an excitation in mode  $a$  is associated to the part of the error state decoded as logical zero  $|0_L\rangle$  while an excitation in mode  $b$  – to the part of the error state decoded as  $|1_L\rangle$ . We leave the state  $|\psi_L\rangle$  arbitrary as the error correction should be agnostic of the logical information. Other important many-body states necessary for the understanding of the scheme are

$$|\Xi_{i_1, \dots, i_c}\rangle = \frac{1}{\sqrt{c}} \left( \sum_{k=1}^c a |e\rangle\langle 1|_{i_k} + b |f\rangle\langle 0|_{i_k} \right) |i_1, \dots, i_c\rangle \quad (28)$$

which involve ground states of both modes and ions' internal states  $|e\rangle$  and  $|f\rangle$ . Assuming the bosonic mode to remain in the ground and first excited states only, the lowest order term of the Hamiltonian  $H_0 + H_2$  (where  $H_0$  is the detuning part of  $H_1$  in Eq. (18)) in this new basis reads

$$\begin{aligned} \tilde{H} = \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \Delta |\Xi_{i_1, \dots, i_c}\rangle \langle \Xi_{i_1, \dots, i_c}| \\ + \delta |i_1, \dots, i_c\rangle \langle i_1, \dots, i_c| \\ + \sqrt{c} G |i_1, \dots, i_c\rangle \langle \Xi_{i_1, \dots, i_c}| + \text{H.c.} \end{aligned} \quad (29)$$

Note that  $\tilde{H}$  is only a portion of the complete Hamiltonian (indicated by the *tilde*) as it does not capture higher order processes involving two or more ions in the excited manifold. However, these processes will be out of resonance as happening at higher rates, e.g., the second order processes involving states with exactly two  $|e/f\rangle$  will happen at rates  $2\Delta$  and  $\sqrt{\binom{c}{2}}G$ . The most significant undesired process among these is the one that deteriorates an initial erroneous state by introducing one additional error. In order to describe this process, we define the complementary many-body states  $|\bar{i}_1, \dots, \bar{i}_c\rangle = X_L |i_1, \dots, i_c\rangle$  with  $X_L$  being the logical Pauli X operator. These states are now associating an excitation in mode  $a$  to the part of the error state decoded as logical one  $|1_L\rangle$  and an excitation in mode  $b$  – to the part of the error state decoded as  $|0_L\rangle$ . The last many-body states that we require are

$$\begin{aligned} |\chi_{i_1, \dots, i_c}\rangle = \frac{1}{\sqrt{n-c}} \\ \left( \sum_{i_k \notin \{i_1, \dots, i_c\}} a |e\rangle\langle 1|_{i_k} + b |f\rangle\langle 0|_{i_k} \right) |\bar{i}_1, \dots, \bar{i}_c\rangle \end{aligned} \quad (30)$$

where the sum is taken over all non erroneous qubits of the state  $|i_1, \dots, i_c\rangle$ . These states are the undesired counterparts of the states  $|\Xi_{i_1, \dots, i_c}\rangle$  defined in Eq. (28). The lowest order undesired Hamiltonian associated to  $H_0 + H_2$

then reads

$$\begin{aligned} \tilde{H}_u = \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \Delta |\chi_{i_1, \dots, i_c}\rangle \langle \chi_{i_1, \dots, i_c}| \\ + \delta |\overline{i_1, \dots, i_c}\rangle \langle \overline{i_1, \dots, i_c}| \\ + \sqrt{n-c} G |\overline{i_1, \dots, i_c}\rangle \langle \chi_{i_1, \dots, i_c}|. \end{aligned} \quad (31)$$

The last term  $H_3$  of the total Hamiltonian  $H$  represents single addressing lasers that couple each ion at position  $j$  to an auxiliary motional mode  $c_j$  which in turn is sympathetically cooled. For a large enough cooling rate, i.e.  $\kappa \gg g$ , we can adiabatically eliminate the auxiliary mode and approximate the whole process with jump operators acting on every ion individually

$$L_{\text{eng},j} = \sqrt{\kappa_{\text{eng}}} (|0\rangle \langle e|_j + |1\rangle \langle f|_j) \quad (32)$$

with  $\kappa_{\text{eng}} \propto g^2/\kappa$ .

We derive the effective Liouvillian that governs the dynamics of the many-body system following the effective operator formalism [80]. We assume that the ground subspace of the many-body Hilbert space is spanned by  $\{|0\rangle, |1\rangle\}^{\otimes n} \otimes |0, 0\rangle$ , while the excited manifold is spanned by states containing at most one excited state, which could be  $|e\rangle$ ,  $|f\rangle$ ,  $|1, 0\rangle$  or  $|0, 1\rangle$ . The non-Hermitian Hamiltonian reads

$$\begin{aligned} H_{\text{NH}} &= H_0 + H_2 + H_3 - \frac{i}{2} \sum_{j=1}^n L_j^\dagger L_j \\ &\approx H_0 + H_2 - \frac{i}{2} \sum_{j=1}^n L_{\text{eng},j}^\dagger L_{\text{eng},j} \\ &\approx \tilde{H} + \tilde{H}_u - \frac{i}{2} \sum_{j=1}^n L_{\text{eng},j}^\dagger L_{\text{eng},j} =: \tilde{H}_{\text{NH}} \end{aligned} \quad (33)$$

where the first approximation uses Eq. (32), while the second approximation follows from considering the lowest order processes of  $H_0 + H_2$  presented in Eqs. (29) and (31). We note that the last term of the expression can be rewritten in terms of  $|\Xi_{i_1, \dots, i_c}\rangle$  and  $|\chi_{i_1, \dots, i_c}\rangle$  in a similar manner as  $H_0$  and results in an energy shift of  $-i\kappa_{\text{eng}}/2$ . Thanks to the block form of  $\tilde{H}_{\text{NH}}$  we invert every block individually

$$\begin{aligned} [\tilde{H}_{\text{NH}}]^{-1} &= \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \frac{1}{\Delta'(c)} |\Xi_{i_1, \dots, i_c}\rangle \langle \Xi_{i_1, \dots, i_c}| \\ &\quad + \frac{1}{\Delta'(n-c)} |\chi_{i_1, \dots, i_c}\rangle \langle \chi_{i_1, \dots, i_c}| + \dots \end{aligned} \quad (34)$$

For convenience, we present here only the relevant part of  $[\tilde{H}_{\text{NH}}]^{-1}$  with the complete rest of the terms found in Supplementary Materials. The scalar parameter in Eq. (34) are defined with the functions

$$\Delta'(x) = \Delta - \frac{i}{2} \kappa_{\text{eng}} - \frac{x G^2}{\delta} \quad (35)$$

where  $x \in \{1, \dots, \ell\}$ . The behavior of these functions in terms of the integer  $x$  will be the core of the error correcting process.

To derive the effective jump operators we must express the drive part  $V := V_+ + V_-$  of the Hamiltonian  $H_1$  in Eq. (18) in terms of our many-body states. In the lowest order, the drive reads

$$\begin{aligned} \tilde{V}_+ = \tilde{V}_-^\dagger &= \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \sqrt{c} \frac{\Omega}{2} |\Xi_{i_1, \dots, i_c}\rangle \langle i_1, \dots, i_c|_0 \\ &\quad + \sqrt{n-c} \frac{\Omega}{2} |\chi_{i_1, \dots, i_c}\rangle \langle i_1, \dots, i_c|_0 \end{aligned} \quad (36)$$

where we define  $|i_1, \dots, i_c\rangle_0$  as the same state  $|i_1, \dots, i_c\rangle$  but with zero excitation in oscillators  $a$  and  $b$ . These states are effectively representing error states of weight  $c$  of the repetition code. We consider this drive to be weak such that one can treat it as a perturbative excitation of the ground subspace of the system. Following Ref. [80], the effective jump operators become

$$\begin{aligned} L_{\text{eff},j} &= L_{\text{eng},j} [\tilde{H}_{\text{NH}}]^{-1} \tilde{V}_+ \\ &= \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_{c-1} \neq j}^n \sqrt{\kappa_{\text{eff}}(c)} |i_1, \dots, i_{c-1}\rangle \langle i_1, \dots, i_c|_0 \\ &\quad + \sum_{i_1 \neq \dots \neq i_c \neq j}^n \sqrt{\kappa_{\text{eff}}(n-c)} |i_1, \dots, i_c, i_{c+1} = j\rangle \langle i_1, \dots, i_c|_0 \end{aligned}$$

These jump operators consist of two processes: a desired one that reduces the weight of error states from  $c$  to  $c-1$ , and an undesired one that increases the error weight by 1. The desired process, represented by the first sum, corrects all error states in which the  $j$ -th ion is erroneous. In contrast, the undesired process flips the state of the  $j$ -th ion even when it is not erroneous, thereby introducing an additional error. Rewriting the jump operator in terms of projectors into the error subspaces, we get the expression in Eq. (21) of the main text. The effective rate function  $\kappa_{\text{eff}}(x)$  is obtained using

$$\kappa_{\text{eff}}(x) = \kappa_{\text{eng}} \frac{\Omega^2}{4} |\Delta'(x)|^{-2} \quad (37)$$

and equals exactly the expression in Eq. (22) of the main text. Fixing the coupling in the Hamiltonian  $H_2$  to  $G = G_0 = \sqrt{\Delta \delta / x_0}$  with an arbitrary  $x_0 \in \{1, \dots, \ell\}$  reduces the previous expression to

$$\kappa_{\text{eff}}(x, x_0) = \frac{\Omega^2}{\kappa_{\text{eng}}} \left[ \left( \frac{2\Delta}{\kappa_{\text{eng}} x_0} \right)^2 (x - x_0)^2 + 1 \right]^{-1}. \quad (38)$$

Both Eqs. (22) and (38) describe a Lorentzian function of the integer parameter  $x$  centered at  $x_0$ . Therefore, choosing the resonance condition  $G = G_0$  favors the desired process  $x_0 \rightarrow x_0 - 1$ , i.e. the correction of error

states of weight  $x_0$  towards errors states of weight  $x_0 - 1$ . The scale and height of the Lorentzian functions are defined as  $x_0 \kappa_{\text{eng}} / 2\Delta$  and  $\Omega^2 / \kappa_{\text{eng}}$ , respectively.

Further discussion about the derivations above is provided in the Supplementary Materials.

## Acknowledgements

I.R. thanks Wojciech Adamczyk, Matteo Simoni, and Jonathan Home for insightful discussions on quantum error correction and its analogy to cooling, as well as for their valuable feedback on the manuscript. The authors acknowledge funding from the Swiss National Science Foundation (Ambizione Grant No. PZ00P2.186040) and the ETH Research Grant ETH-49 20-2.

The project was conceived by F.R. The trapped-ion implementation was developed jointly by I.R. and F.R. I.R. carried out the remaining theoretical derivations. Numerical simulations were performed by I.R. in collaboration with E.Z. F.R. provided guidance throughout the project. All authors contributed to the discussions, interpretation of the results, and writing of the manuscript. The authors declare that they have no competing interests.

- 
- [1] R. Ozeri, arXiv:1310.3432 (2013).
  - [2] I. M. Georgescu, S. Ashhab, and F. Nori, *Rev. Mod. Phys.* **86**, 153 (2014).
  - [3] A. Ekert and C. Macchiavello, *Phys. Rev. Lett.* **77**, 2585 (1999).
  - [4] C. Ryan-Anderson, J. G. Bohnet, K. Lee, D. Gresh, A. Hankin, *et al.*, *Phys. Rev. X* **11**, 041058 (2021).
  - [5] S. Krinner, N. Lacroix, A. Remm, A. Di Paolo, E. Genois, *et al.*, *Nature* **605**, 669 (2022).
  - [6] D. Bluvstein, S. J. Evered, A. A. Geim, S. H. Li, H. Zhou, *et al.*, *Nature* **626**, 58 (2024).
  - [7] R. Acharya, D. A. Abanin, L. Aghababaie-Beni, I. Aleiner, T. I. Andersen, *et al.*, *Nature*, 1 (2024).
  - [8] A. Paetznick, M. P. d. Silva, C. Ryan-Anderson, J. M. Bello-Rivas, J. P. C. III, *et al.*, arXiv:2404.02280 (2024).
  - [9] A. Eickbusch, M. McEwen, V. Sivak, A. Bourassa, J. Atalaya, *et al.*, arXiv:2412.14360 (2024).
  - [10] N. Lacroix, A. Bourassa, F. J. H. Heras, L. M. Zhang, J. Bausch, *et al.*, arXiv:2412.14256 (2024).
  - [11] K. DeBry, N. Meister, A. V. Martinez, C. D. Bruzewicz, X. Shi, *et al.*, arXiv:2503.13908 (2025).
  - [12] Y. Li, Q. Mei, Q.-X. Jie, W. Cai, Y. Li, *et al.*, arXiv:2504.16746 (2025).
  - [13] N. Ofek, A. Petrenko, R. Heeres, P. Reinhold, Z. Leghtas, *et al.*, *Nature* **536**, 441 (2016).
  - [14] L. Hu, Y. Ma, W. Cai, X. Mu, Y. Xu, *et al.*, *Nat. Phys.* **15**, 503 (2019).
  - [15] C. Flühmann, T. L. Nguyen, M. Marinelli, V. Negnevitsky, K. Mehta, and J. P. Home, *Nature* **566**, 513 (2019).
  - [16] S. Rosenblum, P. Reinhold, M. Mirrahimi, L. Jiang, L. Frunzio, and R. J. Schoelkopf, *Science* **361**, 266 (2018).
  - [17] D. J. Reilly, in *2019 IEEE International Electron Devices Meeting (IEDM)* (2019) pp. 31.7.1–31.7.6, ISSN: 2156-017X.
  - [18] P. Campagne-Ibarcq, A. Eickbusch, S. Touzard, E. Zolys-Geller, N. E. Frattini, *et al.*, *Nature* **584**, 368 (2020).
  - [19] B. M. Terhal, *Rev. Mod. Phys.* **87**, 307 (2015).
  - [20] J. P. Paz and W. H. Zurek, *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences* **454**, 355 (1998).
  - [21] C. Ahn, A. C. Doherty, and A. J. Landahl, *Phys. Rev. A* **65**, 042301 (2002).
  - [22] M. Sarovar and G. J. Milburn, *Phys. Rev. A* **72**, 012306 (2005).
  - [23] O. Oreshkov, “Continuous-time quantum error correction,” in *Quantum Error Correction*, edited by D. A. Lidar and T. A. Brun (Cambridge University Press, Cambridge, United Kingdom, 2013) Chap. 8, pp. 201–228.
  - [24] J. Lebreuilly, K. Noh, C.-H. Wang, S. M. Girvin, and L. Jiang, arXiv:quant-ph/2103.05007 (2021), 2103.05007.
  - [25] D. Gottesman, A. Kitaev, and J. Preskill, *Phys. Rev. A* **64**, 012310 (2001).
  - [26] M. Mirrahimi, Z. Leghtas, V. V. Albert, S. Touzard, R. J. Schoelkopf, *et al.*, *New J. Phys.* **16**, 045014 (2014).
  - [27] M. H. Michael, M. Silveri, R. Brierley, V. V. Albert, J. Salmilehto, *et al.*, *Phys. Rev. X* **6**, 031006 (2016).
  - [28] B. de Neeve, T.-L. Nguyen, T. Behrle, and J. P. Home, *Nat. Phys.*, 1 (2022).
  - [29] Z. Leghtas, S. Touzard, I. M. Pop, A. Kou, B. Vlastakis, *et al.*, *Science* **347**, 853 (2015).
  - [30] S. Touzard, A. Grimm, Z. Leghtas, S. Mundhada, P. Reinhold, *et al.*, *Phys. Rev. X* **8**, 021005 (2018).
  - [31] A. Grimm, N. E. Frattini, S. Puri, S. O. Mundhada, S. Touzard, *et al.*, *Nature* **584**, 205 (2020).
  - [32] R. Lescanne, M. Villiers, T. Peronnin, A. Sarlette, M. Delbecq, *et al.*, *Nat. Phys.* **16**, 509 (2020).
  - [33] J. M. Gertler, B. Baker, J. Li, S. Shirol, J. Koch, and C. Wang, *Nature* **590**, 243 (2021).
  - [34] R. Rousseau, D. Ruiz, E. Albertinale, P. d’Avezac, D. Banys, *et al.*, arXiv:2502.07892 (2025).
  - [35] V. V. Sivak, A. Eickbusch, B. Royer, S. Singh, I. Tsioutsios, *et al.*, *Nature* **616**, 50 (2023).
  - [36] Z. Ni, S. Li, X. Deng, Y. Cai, L. Zhang, *et al.*, *Nature* **616**, 56 (2023).
  - [37] H. Putterman, K. Noh, C. T. Hann, G. S. MacCabe, S. Aghaieimebodi, *et al.*, *Nature* **638**, 927 (2025).
  - [38] J. Kerckhoff, H. I. Nurdin, D. S. Pavlichin, and H. Mabuchi, *Phys. Rev. Lett.* **105**, 040502 (2010).
  - [39] F. Pastawski, L. Clemente, and J. I. Cirac, *Phys. Rev. A* **83**, 012304 (2011).
  - [40] J. Cohen and M. Mirrahimi, *Phys. Rev. A* **90**, 062344 (2014).
  - [41] M. Ippoliti, L. Mazza, M. Rizzi, and V. Giovannetti, *Phys. Rev. A* **91**, 042322 (2015).
  - [42] E. Kapit, *Phys. Rev. Lett.* **116**, 150501 (2016).
  - [43] K.-C. Hsu and T. A. Brun, *Phys. Rev. A* **93**, 022321 (2016).
  - [44] F. Reiter, A. S. Sørensen, P. Zoller, and C. A. Muschik, *Nat. Commun.* **8**, 1822 (2017).
  - [45] J.-M. Lihm, K. Noh, and U. R. Fischer, *Phys. Rev. A* **98**, 012317 (2018).
  - [46] S. Heußen, D. F. Locher, and M. Müller, *PRX Quantum* **5**, 010333 (2024).
  - [47] W. P. Livingston, M. S. Blok, E. Flurin, J. Dressel, A. N. Jordan, and I. Siddiqi, *Nat. Commun.* **13**, 2307 (2022).

- [48] Z. Li, T. Roy, D. Rodríguez Pérez, K.-H. Lee, E. Kapit, and D. I. Schuster, *Nat. Commun.* **15**, 1681 (2024).
- [49] O. Shtanko, Y.-J. Liu, S. Lieu, A. V. Gorshkov, and V. V. Albert, “Bounds on Autonomous Quantum Error Correction,” (2023).
- [50] E. Knill and R. Laflamme, *Phys. Rev. A* **55**, 900 (1997).
- [51] P. Aliferis and J. Preskill, *Phys. Rev. A* **78**, 052331 (2008).
- [52] D. W. Leung, M. A. Nielsen, I. L. Chuang, and Y. Yamamoto, *Phys. Rev. A* **56**, 2567 (1997).
- [53] J. Dalibard, Y. Castin, and K. Mølmer, *Phys. Rev. Lett.* **68**, 580 (1992).
- [54] H. J. Carmichael, *Statistical Methods in Quantum Optics 2: Non-classical fields*, Statistical methods in quantum optics No. H. J. Carmichael ; 2 (Springer, Berlin, 2008).
- [55] K. Mølmer, Y. Castin, and J. Dalibard, *Journal of the Optical Society of America B* **10**, 524 (1993).
- [56] T. E. O’Brien, B. Tarasinski, and L. DiCarlo, *npj Quantum Inf.* **3**, 1 (2017).
- [57] M. W. van Mourik, E. Zapusek, P. Hrmo, L. Gerster, R. Blatt, *et al.*, *Phys. Rev. Lett.* **132**, 040602 (2024).
- [58] O. Higgott and C. Gidney, *Quantum* **9**, 1600 (2025).
- [59] J. Edmonds, *Journal of Research of the National Bureau of Standards Section B Mathematics and Mathematical Physics* **69B**, 125 (1965).
- [60] J. Edmonds, *Can. J. Math.* **17**, 449 (1965).
- [61] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, *J. Math. Phys.* **43**, 4452 (2002).
- [62] A. G. Fowler, A. C. Whiteside, and L. C. L. Hollenberg, *Phys. Rev. Lett.* **108**, 180501 (2012).
- [63] A. d. iOlius, P. Fuentes, R. Orús, P. M. Crespo, and J. E. Martinez, *Quantum* **8**, 1498 (2024).
- [64] C. Jones, *arXiv:2408.12135* (2024).
- [65] K. R. Ott, B. Hetényi, and M. E. Beverland, *arXiv:2502.16408* (2025).
- [66] L. A. Beni, O. Higgott, and N. Shutty, *arXiv:2503.10988* (2025).
- [67] G. Pantaleoni, B. Q. Baragiola, and N. C. Menicucci, *Phys. Rev. Lett.* **125**, 040501 (2020).
- [68] L. Gilles, B. M. Garraway, and P. L. Knight, *Phys. Rev. A* **49**, 2785 (1994).
- [69] C. Chamberland, K. Noh, P. Arrangoiz-Arriola, E. T. Campbell, C. T. Hann, *et al.*, *PRX Quantum* **3**, 010329 (2022).
- [70] I. Rojko, M. Simoni, E. Zapusek, F. Reiter, and J. Home, *arXiv:quant-ph/2407.18087* (2024).
- [71] B. Royer, S. Singh, and S. M. Girvin, *Phys. Rev. Lett.* **125**, 260509 (2020).
- [72] M. H. Shaw, A. C. Doherty, and A. L. Grimsmo, *PRX Quantum* **5**, 010331 (2024).
- [73] B. de Neeve, T.-L. Nguyen, A. Ferk, T. Behrle, F. Lancelotti, *et al.*, *Physical Review Letters* **134**, 203603 (2025).
- [74] C. Siegle and P. Campagne-Ibarcq, *Phys. Rev. A* **108**, 042427 (2023).
- [75] N. P. Breuckmann and J. N. Eberhardt, *PRX Quantum* **2**, 040101 (2021).
- [76] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, *Nature* **627**, 778 (2024).
- [77] I. Rojko, P. M. Röggl, M. Wagener, M. Fontboté-Schmidt, S. Welte, *et al.*, *Phys. Rev. Lett.* **133**, 100601 (2024).
- [78] M. Raghunandan, F. Wolf, C. Ospelkaus, P. O. Schmidt, and H. Weimer, *Science Advances* **6**, eaaw9268 (2020).
- [79] E. Zapusek, I. Rojko, and F. Reiter, *arXiv:2507.02043* (2025).
- [80] F. Reiter and A. S. Sørensen, *Phys. Rev. A* **85**, 032111 (2012).

# Supplementary Materials

Ivan Rojko<sup>1,2</sup>, Elias Zapusek<sup>1,2</sup> and Florentin Reiter<sup>1,2,3</sup>

<sup>1</sup>*Institute for Quantum Electronics, ETH Zürich, Otto-Stern-Weg 1, 8093 Zürich, Switzerland*

<sup>2</sup>*Quantum Center, ETH Zürich, 8093 Zürich, Switzerland*

<sup>3</sup>*Fraunhofer Institute for Applied Solid State Physics IAF, Tullastr. 72, 79108 Freiburg, Germany*

(Dated: July 18, 2025)

## I. TRAPPED-ION IMPLEMENTATION OF DISSIPATIVE REPETITION CODE

Following our derivation from the “Materials and Methods” section, we now present the further details of the trapped-ion implementation of the dissipative repetition code. We start directly with the inversion of the non-Hermitian Hamiltonian  $\tilde{H}_{\text{NH}}$  in Eq. (34) which in the lowest order reads

$$\begin{aligned} [\tilde{H}_{\text{NH}}]^{-1} = & \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \frac{1}{\Delta'(c)} |\Xi_{i_1, \dots, i_c}\rangle \langle \Xi_{i_1, \dots, i_c}| + \frac{1}{\Delta'(n-c)} |\chi_{i_1, \dots, i_c}\rangle \langle \chi_{i_1, \dots, i_c}| + \\ & + \frac{1}{\delta'(c)} |i_1, \dots, i_c\rangle \langle i_1, \dots, i_c| + \frac{1}{\delta'(n-c)} |\overline{i_1, \dots, i_c}\rangle \langle \overline{i_1, \dots, i_c}| + \\ & + \frac{1}{G'(c)} |i_1, \dots, i_c\rangle \langle \Xi_{i_1, \dots, i_c}| + \frac{1}{G'(n-c)} |\overline{i_1, \dots, i_c}\rangle \langle \chi_{i_1, \dots, i_c}| + \text{H.c.} \end{aligned} \quad (1)$$

The scalar parameters in Eq. (1) are defined with the functions

$$\Delta'(x) = \Delta - \frac{i}{2} \kappa_{\text{eng}} - \frac{x G^2}{\delta} \quad \delta'(x) = \delta - \frac{x G^2}{\Delta - \frac{i}{2} \kappa_{\text{eng}}} \quad G'(x) = \sqrt{x} G \left( 1 - \delta \frac{\Delta - \frac{i}{2} \kappa_{\text{eng}}}{x G^2} \right) \quad (2)$$

where  $x \in \{1, \dots, \ell\}$ .

To derive the effective jump operators we must express the drive part  $V := V_+ + V_-$  of the Hamiltonian  $H_1$  in Eq. (18) in terms of our many-body states. In the lowest order, the drive reads

$$V = V_+ + V_- \quad \text{with} \quad \tilde{V}_+ = \tilde{V}_-^\dagger = \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c}^n \sqrt{c} \frac{\Omega}{2} |\Xi_{i_1, \dots, i_c}\rangle \langle i_1, \dots, i_c|_0 + \sqrt{n-c} \frac{\Omega}{2} |\chi_{i_1, \dots, i_c}\rangle \langle i_1, \dots, i_c|_0 \quad (3)$$

where we define  $|i_1, \dots, i_c\rangle_0$  as the same state  $|i_1, \dots, i_c\rangle$  but with zero excitation in oscillators  $a$  and  $b$ . These states are effectively representing error states of weight  $c$  of the repetition code. We consider this drive to be weak such that one can treat it as a perturbative excitation of the ground subspace of the system. Following Ref. [80], the effective jump operators become

$$\begin{aligned} L_{\text{eff},j} = L_{\text{eng},j} [\tilde{H}_{\text{NH}}]^{-1} \tilde{V}_+ = & \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_{c-1} \neq j}^n \sqrt{\kappa_{\text{eng}}} \frac{\Omega}{2} \frac{1}{\Delta'(c)} \underbrace{|i_1, \dots, i_{c-1}\rangle_0}_{(c-1) \text{ errors}} \underbrace{\langle i_1, \dots, i_c|_0}_{(c) \text{ errors}} + \\ & + \sum_{c=1}^{\ell} \sum_{i_1 \neq \dots \neq i_c \neq j}^n \sqrt{\kappa_{\text{eng}}} \frac{\Omega}{2} \frac{1}{\Delta'(n-c)} \underbrace{|i_1, \dots, i_c, i_{c+1} = j\rangle_0}_{(c+1) \text{ errors}} \underbrace{\langle i_1, \dots, i_c|_0}_{(c) \text{ errors}}. \end{aligned} \quad (4)$$

These jump operators consist of two processes: a desired one that reduces the weight of error states from  $c$  to  $c-1$ , and an undesired one that increases the error weight by 1. The desired process, represented by the first sum, corrects all error states in which the  $j$ -th ion is erroneous. In contrast, the undesired process flips the state of the  $j$ -th ion even when it is not erroneous, thereby introducing an additional error. Rewriting the jump operator in terms of projectors into the error subspaces of weight  $c$  gives

$$L_{\text{eff},j} = \sum_{c=1}^{\ell} \sqrt{\kappa_{\text{eff}}(c)} \left( |0\rangle\langle 1|_j P_{1,c} + |1\rangle\langle 0|_j P_{0,c} \right) + \sqrt{\kappa_{\text{eff}}(n-c)} \left( |1\rangle\langle 0|_j P_{1,c} + |0\rangle\langle 1|_j P_{0,c} \right) \quad (5)$$

where the projectors  $P_{0,c}$  and  $P_{1,c}$  are defined as

$$P_{\psi,c} := \sum_{i_i \neq \dots \neq i_c}^n \left( \bigotimes_{k=1}^c X_{i_k} \right) |\psi\rangle\langle\psi|_L \left( \bigotimes_{k=1}^c X_{i_k} \right). \quad (6)$$

The effective rate function  $\kappa_{\text{eff}}(x)$  is obtained using

$$\kappa_{\text{eff}}(x) = \kappa_{\text{eng}} \frac{\Omega^2}{4} |\Delta'(x)|^{-2} = \kappa_{\text{eng}} \frac{\Omega^2}{4} \left[ \left( \Delta - \frac{xG}{\delta} \right)^2 + \frac{1}{4} \kappa_{\text{eng}}^2 \right]^{-1} \quad (7)$$

Fixing the coupling in the Hamiltonian  $H_2$  to  $G = G_0 = \sqrt{\Delta\delta/x_0}$  with an arbitrary  $x_0 \in \{1, \dots, \ell\}$  reduces the previous expression to

$$\kappa_{\text{eff}}(x, x_0) = \frac{\Omega^2}{\kappa_{\text{eng}}} \left[ \left( \frac{2\Delta}{\kappa_{\text{eng}} x_0} \right)^2 (x - x_0)^2 + 1 \right]^{-1}. \quad (8)$$

Both Eqs. (7) and (8) describe a Lorentzian function of the integer parameter  $x$  centered at  $x_0$ . Therefore, choosing the resonance condition  $G = G_0$  favors the desired process  $x_0 \rightarrow x_0 - 1$ , i.e. the correction of error states of weight  $x_0$  towards error states of weight  $x_0 - 1$ . The scale and height of the Lorentzian functions are defined as  $x_0 \kappa_{\text{eng}} / 2\Delta$  and  $\Omega^2 / \kappa_{\text{eng}}$ , respectively. Fig. 6 shows an example of the normalized desired and undesired effective rates for a chain of 51 ions (i.e. repetition code  $[[51, 1, 25]]$ ) and  $x_0 = 10$ . The shaded area represents system states that have more errors than the code distance and are taken into account in undesired process. Although irrelevant for the later discussion, it is interesting to note that as expected the effective rates of the undesired process (red bars in the unshaded area) simply corresponds to the reflection of the Lorentzian function of the desired process past the code distance  $d$  (green bars in the unshaded area).

The whole analysis above did not take into account one last undesired term in the the first order expansion  $\tilde{H}_u$  given in Eq. (31), which is the process that adds a single error to the logical states themselves. Fortunately, this process can be easily included if we allow the abuse of notation  $|i_1, \dots, i_c\rangle$  with  $c = 0$  to represent the logical states  $|0_L\rangle$  and  $|1_L\rangle$ . In this case the effective jump operator representing the  $j$ -th ion will be identical to the expressions in Eqs. (4) and (5) but with the sum of the undesired process starting at  $c = 0$ .

The most important feature of the proposed dissipative system is that every jump operator  $L_{\text{eff},j}$  with rates resonant at  $x_0$  does not correct a unique error state as in standard measurement based stabilizer codes. Instead, it corrects all the error states of weight  $x_0$  involving an error on qubit  $j$ . Another interesting feature is that by fixing the interaction strength  $G = G_0$ , we also correct error states in the neighborhood of  $x_0$ . It remains an open question whether one can engineer resonances of more general shapes which are not Lorentzian. This would allow to correct error states of various weights at the same rate and with even less jump operators. Such processes may be achievable using engineered nonlinearities in the shapes of the global laser drives.

As mentioned above the scale of the Lorentzian is given by  $x_0 \kappa_{\text{eng}} / 2\Delta$ . We can decrease it, and thus decrease the rate of the undesired process, by increasing the internal states' detuning  $\Delta$  or equivalently by increasing the detuning of the oscillators  $\delta$  (because of the condition  $x_0 = \Delta\delta/G^2$ ).

FIG. 6: Normalized rates of the effective jump operators from Eq. (8). Desired and undesired processes follow  $\kappa_{\text{eff}}(c)$  and  $\kappa_{\text{eff}}(n - c)$ , respectively, for  $c \in \{1, \dots, \ell\}$ , shown as green circles and red crosses. The shaded area lies outside the validity range and is included for illustration (see main text). The plot corresponds to the  $[[51, 1, 25]]$  repetition code with parameters:  $\kappa_{\text{eng}} = 225\Gamma$ ,  $\Delta = 15\Gamma$ ,  $G = 120\Gamma$ ,  $\delta = 480\Gamma$ , and  $\Gamma = 1/150$ , which sets the scale of the unnormalized rates.

